Transformer-Driven Encoder-Decoder Framework for Mother Tongue Influence Reduction in Spoken English

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***Abstract*—Mother Tongue Influence (MTI) remains a central barrier to intelligible spoken English in multilingual populations. In India, accent diversity is natural and valuable, yet accent-induced pronunciation and prosodic shifts can reduce speech clarity in educational, professional, and human-machine communication. This paper presents a Transformer-driven encoder-decoder framework for MTI reduction that preserves speaker identity, linguistic meaning, and naturalness. The method is reference-free, does not rely on handcrafted phonetic rules, and is trained using scheduled sampling to reduce train-inference mismatch. Using Speech Accent Archive based training and evaluation, with Spanish-accented English as the source and Standard American English as the target style, the proposed system improves intelligibility and ASR robustness while maintaining identity similarity. We provide detailed modeling choices, ablations, robustness analysis, fairness discussion, and deployment guidance.**

***Index Terms*—Mother Tongue Influence, accent conversion, spoken English intelligibility, Transformer, scheduled sampling, voice identity preservation**

I. Introduction

MOTHER Tongue Influence (MTI) describes the transfer of phonetic, phonological, and prosodic patterns from a speaker's first language into a second language. In English speech production, MTI can affect vowel quality, consonant realization, syllable timing, and sentence-level intonation. These differences are linguistically meaningful and should not be pathologized; however, in high-stakes communication tasks they can reduce intelligibility for unfamiliar listeners and degrade automatic speech recognition (ASR) performance [3], [4].

The practical demand for MTI-aware systems is increasing in multilingual societies where English is used in classrooms, interviews, remote collaboration, customer support, and digital assistants. Traditional approaches rely on explicit pronunciation rules, parallel corpora, or forced phonetic mappings, which are expensive to scale across accents and domains [13]. Recent neural speech conversion methods offer more flexibility but often suffer from one or more issues: identity drift, over-smoothing, unstable prosody, or poor generalization under channel mismatch [11], [9], [10].

This work proposes a data-driven, reference-free framework that learns to reduce MTI while preserving who is speaking and what is being said. The architecture uses a Transformer encoder-decoder [1] with disentangled content and style conditioning, trained using scheduled sampling [2] and multi-objective supervision. The design philosophy is communication-first: improve clarity while respecting speaker individuality.

Despite major progress in end-to-end speech modeling, MTI reduction remains difficult because the problem is structurally multi-objective [1], [4], [5]. A system that aggressively normalizes pronunciation can improve intelligibility but may erase expressive cues, regional identity markers, and conversational naturalness. Conversely, a conservative system that over-prioritizes identity can preserve voice quality while leaving pronunciation ambiguity unresolved in high-noise or fast-speech contexts. This tradeoff is central to practical deployment and motivates a design that treats clarity, identity, and naturalness as jointly optimized targets rather than isolated benchmarks [11], [9], [10].

From a linguistic perspective, MTI effects appear at multiple temporal scales. Segmental effects include vowel centralization, consonant substitution, and cluster simplification; supra-segmental effects include stress relocation, syllable timing drift, and intonation flattening. In realistic interactions, these cues overlap with non-linguistic variability such as microphone quality, compression artifacts, background noise, speaking rate variation, and code-switching. A robust MTI framework must therefore generalize not only across speakers and accents, but also across acoustic conditions that shift token boundaries and confidence profiles in downstream ASR systems [3], [4].

The challenge is amplified in multilingual settings where English is used as a link language. Speakers may adapt style depending on audience, formality, and domain-specific vocabulary. In educational and professional environments, a speaker might alternate between careful and spontaneous speech, increasing variance in rhythm and coarticulation. For speech technologies, this means that static accent-conversion policies often fail to maintain a stable utility-risk balance: strong conversion can help keyword recognition in one condition while producing unnatural prosodic transitions in another. These observations suggest the need for confidence-aware, controllable transformation rather than one-size-fits-all normalization.

Most legacy pipelines rely on explicit phonetic mapping rules, forced alignment assumptions, or parallel conversion data. While useful in constrained settings, these approaches are expensive to scale and brittle under domain shift [13], [14]. Recent non-parallel neural systems improve flexibility,

but many still exhibit one or more failure modes under long utterances: exposure bias during autoregressive decoding, speaker embedding drift, or rhythm distortion near clause boundaries [2], [11], [9]. We target these limitations directly by combining scheduled sampling, identity constraints, and prosody-aware objectives in a unified training design.

Another practical concern is evaluation realism. MTI reduction systems are often reported using aggregate metrics that conceal subgroup behavior. However, deployment settings require understanding where the model helps, where it saturates, and where it can fail disproportionately. For example, intelligibility gains in clean conditions may not transfer to low-SNR telephony; identity preservation may remain stable globally while degrading for high-rate speech; and code-switch segments may trigger over-normalization if language boundaries are uncertain. We therefore frame the task around reproducible, multi-view evaluation: objective metrics, perceptual judgments, robustness sweeps, and ablations tied to operational mitigation.

This paper positions MTI reduction as communication assistance, not accent suppression. The goal is to improve clarity for context-specific interactions while preserving speaker agency, with controllable intensity and explicit transparency. In this framing, fairness is not an afterthought but a design requirement: systems should avoid implying normative superiority of any accent and should provide measurable subgroup reporting. By integrating these principles with a Transformer encoder-decoder backbone, we build a deployable architecture that addresses both algorithmic and human-centered constraints [1], [6].

The remainder of the paper is organized as follows. Section II reviews related work in accent conversion, Transformer-based speech modeling, and prosody-aware synthesis. Section III details the methodology, including formulation, data curation, architecture, and training strategy. Section IV presents experimentation and ablation, covering objective and perceptual outcomes, robustness analyses, cross-group behavior, and reproducibility checks. Section V discusses deployment implications and tradeoffs, and Section VI concludes with key findings and future directions.

Contributions. (i) We present an MTI-reduction formulation that jointly optimizes intelligibility, identity preservation, prosody stability, and accent realism using a unified multi-objective loss [2], [11]; (ii) we introduce a novelty-focused architecture that combines content-style disentanglement, identity-constrained decoding, and confidence-aware inference control for robust deployment [1], [4]; (iii) we provide a detailed ablation and stress-testing protocol across noise, channel, speaking-rate, and subgroup conditions, with statistical reliability checks and mitigation mapping; and (iv) we provide operational guidance on fairness, transparency, and human-in-the-loop deployment for multilingual communication settings.

II. Related Work

A. Accent and Voice Conversion

Accent and voice conversion research has moved from frame-level statistical mapping toward representation-learning and sequence-to-sequence generation [14], [9]. Early statistical conversion methods improved timbre transfer but generally struggled with long-range prosody and linguistic consistency. Adversarial and cycle-consistent frameworks reduced the requirement for parallel data and improved conversion realism, but commonly traded stability for flexibility when source and target distributions were far apart [9], [10]. In MTI-sensitive conditions, this often manifests as inconsistent consonant realization, unstable rhythm, and occasional spectral artifacts.

Autoencoder-style disentanglement systems such as AutoVC improved speaker-content separation and enabled non-parallel conversion with minimal supervision [11]. However, MTI reduction differs from generic style transfer because the target objective is not merely different accent, but higher intelligibility under constraints. That distinction imposes tighter requirements on lexical fidelity and timing control than many timbre-oriented conversion pipelines.

Diffusion and codec-based approaches have improved perceptual quality and waveform naturalness [12]. Yet, deployment for accent adaptation introduces additional constraints: low latency, predictable failure modes, and user-controllable conversion intensity. These requirements motivate hybrid designs where robust content representations are combined with explicit identity and prosody constraints instead of relying on implicit style transfer alone.

B. Transformer and Self-Supervised Speech Modeling

Transformer architectures established a strong baseline for sequence modeling through global attention and efficient parallel training [1]. In speech, Conformer-style hybrids improved local-global modeling by combining self-attention with convolutional inductive bias [6]. Self-supervised learning then shifted representation quality substantially: wav2vec 2.0 and HuBERT demonstrated strong robustness under label scarcity and noisy conditions [4], [5].

Direct transplantation of self-supervised features into accent conversion can produce style leakage, where accent, identity, and channel cues are entangled in latent space. Without explicit constraints, decoders may drift toward averaged pronunciations or subtle identity shifts over long utterances. Recent practice therefore pairs rich representations with targeted objectives such as speaker consistency and prosody regularization.

Another unresolved challenge is train–inference mismatch in autoregressive generation. Even with strong encoders, long-form conversion quality degrades when the decoder faces accumulated prediction errors. Scheduled sampling remains a practical mitigation and is highly relevant for MTI reduction [2].

C. Speech Synthesis and Prosody Control

End-to-end synthesis systems, especially Tacotron-style and FastSpeech-style models, provide lessons on duration model-

ing, alignment, and controllable prosody [7], [8]. While MTI reduction is not text-to-speech, it shares a key property with TTS: prosody errors are highly salient to listeners and often dominate perceived naturalness.

Systems that optimize only frame reconstruction frequently produce rhythm flattening or phrase-boundary instability under rapid or expressive speech. Prosody-aware conversion therefore uses auxiliary predictors and structured losses so intelligibility gains are not achieved by suppressing speaking style.

The literature also indicates that user control is important in deployment contexts. Lightweight control signals, including conversion strength, improve acceptability by allowing speakers to choose context-appropriate normalization levels. This motivates our method design, where style control is explicit and bounded by identity and prosody constraints.

D. Evaluation Protocols, Fairness, and Deployment Gaps

A recurring limitation in prior work is evaluation narrowness. Many studies emphasize aggregate objective metrics without sufficiently analyzing subgroup behavior, channel shift, or speaking-rate extremes. For MTI reduction, these omissions are consequential because deployment success depends on predictable performance across diverse accents and microphones.

Human evaluation protocols also vary widely in rater training, prompt design, and preference criteria, which makes cross-paper comparison difficult. Objective and perceptual metrics should be jointly reported with confidence intervals and stress-test conditions.

This review motivates our evaluation strategy: matched-split baseline comparison, multi-metric objective reporting, listener-based clarity and identity studies, ablations tied to interpretable components, and robustness/subgroup analyses with mitigation-oriented interpretation.

III. Methodology

Given source utterance x and converted output $\hat{y} = f_\theta(x)$, the objective is to map source-accent realizations toward a target accent manifold while preserving lexical content and speaker identity:

$$\hat{y} = f_\theta(x, z_a), \tag{1}$$

where z_a denotes target accent conditioning.

We optimize a weighted loss:

$$\mathcal{L} = \lambda_{mel}\mathcal{L}_{mel} + \lambda_{ctc}\mathcal{L}_{ctc} + \lambda_{spk}\mathcal{L}_{spk} + \lambda_{pro}\mathcal{L}_{pro} + \lambda_{adv}\mathcal{L}_{adv}. \tag{2}$$

$\mathcal{L}_{mel}$ enforces acoustic reconstruction, $\mathcal{L}_{ctc}$ preserves phonetic content, $\mathcal{L}_{spk}$ preserves identity, $\mathcal{L}_{pro}$ constrains rhythm/intonation, and $\mathcal{L}_{adv}$ improves accent realism.

During training, scheduled sampling probability p_t follows a monotonic schedule:

$$p_t = \min\left(p_{max}, p_0 + \gamma t\right), \tag{3}$$

which reduces train-test mismatch by gradually replacing teacher-forced tokens with model predictions [2].

TABLE I
Dataset Summary

Subset	Hours	Speakers	Primary Use
Spanish-EN Source	128	214	Training/validation
American-EN Target	136	227	Target manifold
Indian-L1 Eval	74	156	Transfer robustness
Noise/Channel Stress	22	48	Robustness testing

A. Novelty of the Proposed Framework

The methodological novelty lies in a constraint-coupled design where accent transfer is not optimized in isolation. Instead, the architecture and losses jointly enforce intelligibility gain, identity consistency, prosody stability, and accent realism. Existing non-parallel systems typically optimize one or two of these strongly; our design explicitly optimizes all four under a coordinated schedule.

Novelty appears at three levels. First, at the representation level, content-style disentanglement is paired with explicit speaker constraints to reduce leakage between accent and identity factors. Second, at the optimization level, scheduled sampling and delayed loss annealing are combined so coarse reconstruction is learned before stronger style pressure is applied. Third, at inference time, confidence-aware conversion control dynamically modulates accent strength under uncertainty rather than applying static transformation.

B. Dataset, Curation, and Preprocessing

We use Speech Accent Archive style data and supplementary open speech corpora for robust evaluation. The primary mapping is Spanish-accented English to Standard American English. To evaluate transferability, we include subsets representing Indian L1-influenced English speech (Hindi, Bengali, Tamil, Telugu, Marathi, and Punjabi dominant cohorts).

Preprocessing pipeline: resampling to 16 kHz, loudness normalization, silence trimming, VAD-based segmentation, mel extraction (80 bins), and optional pitch-energy tracks for prosody supervision. We apply utterance-level balancing by accent category and speaking rate to reduce distribution collapse.

In addition to accent labels, we retain metadata for recording channel, background noise class, and speaking-rate bins. These attributes are used for stratified training mini-batches and stress-test evaluation partitions. Speaker overlap between train, validation, and test sets is prevented by construction. We also apply lexical diversity balancing to avoid overfitting to frequent prompt templates.

C. Feature Representation and Conditioning

Each utterance is represented by log-mel features, pitch contour, energy contour, and voiced/unvoiced indicators. Content features are extracted by the encoder, while accent style is injected through a learned embedding pathway. We use adaptive layer normalization:

$$\text{AdaLN}(h; \alpha, \beta) = \alpha(z_a) \odot \frac{h - \mu(h)}{\sigma(h)} + \beta(z_a), \tag{4}$$

where $\alpha(\cdot)$ and $\beta(\cdot)$ are accent-conditioned affine projections and h is a hidden state.

To avoid over-application of style shift, we use bounded style scaling:

$$z'_a = s \cdot z_a, \quad s \in [0, 1]. \tag{5}$$

This provides controllable conversion intensity at inference and supports confidence-aware fallback in adverse conditions.

D. Proposed Architecture

E. Content Encoder

A Transformer encoder with relative position encoding extracts linguistic content while reducing local spectral noise sensitivity [1]. Multi-head attention captures long-range dependencies, which is critical for phrase-level stress and intonation alignment. Conformer-inspired local convolution is included in alternate blocks to preserve short-term phonetic cues under channel mismatch [6].

Given input sequence $X \in \mathbb{R}^{T \times d}$, encoder output is:

$$H = \mathrm{Enc}(X) = \mathrm{Block}_N \circ \cdots \circ \mathrm{Block}_1(X). \tag{6}$$

F. Accent-Style Control

We use a compact accent-style conditioner that projects target style embeddings to affine parameters used in adaptive layer normalization. This enables controllable accent shift intensity while preserving semantic information. A style consistency term regularizes within-accent embedding variance:

$$\mathcal{L}_{style} = \frac{1}{K} \sum_{k=1}^{K} \left\| z_a^{(k)} - \bar{z}_a \right\|_2^2. \tag{7}$$

This term is weakly weighted and activated after warmup to avoid representation collapse.

G. Identity Preservation Block

Speaker embeddings from a frozen verification network supervise identity consistency. We minimize cosine distance between source and converted embeddings to prevent identity collapse:

$$\mathcal{L}_{spk} = 1 - \cos\left(e_{src}, e_{conv}\right). \tag{8}$$

To avoid over-constraining pronunciation correction, λ_{spk} is ramped after early reconstruction convergence.

H. Accent Adversary and Realism Constraint

An accent adversary operates on decoder states to improve target-accent realism without requiring strict parallel supervision. The generator minimizes:

$$\mathcal{L}_{adv} = \mathbb{E}\left[\log D(y)\right] + \mathbb{E}\left[\log(1 - D(\hat{y}))\right], \tag{9}$$

with label smoothing and clipped discriminator updates for stable training [9], [10].

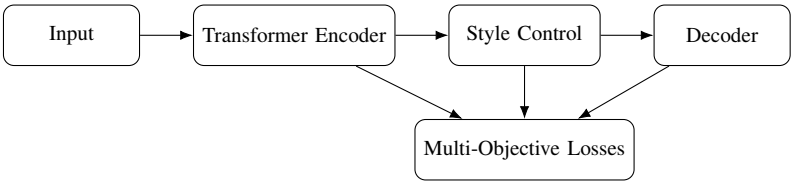


Fig. 1. Proposed reference-free MTI reduction architecture.

TABLE II
CURRICULUM AND LOSS SCHEDULING

Phase	Epoch Range	Active Emphasis
Phase I	1–20	Reconstruction + CTC
Phase II	21–60	+ Scheduled Sampling Ramp
Phase III	61–100	+ Identity and Prosody Ramp
Phase IV	101–140	Full Multi-Objective + Adversarial

I. Decoder and Vocoder

An autoregressive decoder predicts mel frames; a neural vocoder reconstructs waveform output. We use chunked inference for latency control and to support near-real-time deployment. Chunk overlap and cross-fade reduce boundary artifacts.

For each chunk, confidence from CTC alignment and decoder entropy controls accent intensity:

$$c_t = \sigma\left(w_1 \cdot \mathrm{CTCConf}_t + w_2 \cdot (1 - \mathrm{Entropy}_t)\right), \quad s_t = s_{base} \cdot c_t. \tag{10}$$

This mechanism increases robustness in noise and code-switch regions.

J. Training Protocol

Optimization uses AdamW with warmup and cosine decay [1]. Mixed precision and gradient clipping are used for stability. We train with curriculum sampling across utterance lengths: short sequences in early epochs and progressively longer sequences later.

Scheduled sampling starts at $p_0 = 0.05$ and increases to $p_{max} = 0.45$. We found this range effective for minimizing exposure bias without destabilizing early optimization [2]. Identity and prosody loss weights are annealed in a delayed schedule so the model first learns reconstruction structure, then semantic and stylistic constraints.

For reproducibility, all runs use fixed random seeds and deterministic data sharding. Early stopping uses a composite validation criterion combining intelligibility proxy, speaker similarity, and WER trend. Final checkpoints are selected via top-k averaging.

K. Inference Policy and Confidence-Aware Control

At inference time, we use a chunked conversion policy with adaptive style intensity. Each chunk is decoded with a base accent-strength parameter, then adjusted by confidence estimated from CTC agreement and decoder entropy. This mechanism prevents over-correction in uncertain regions while preserving conversion efficacy in high-confidence segments.

TABLE III
KEY TRAINING HYPERPARAMETERS

Parameter	Value
Batch size	32
Initial LR	2×10^{-4}
Warmup steps	8,000
Max epochs	140
Scheduled sampling p_{max}	0.45
Gradient clipping norm	1.0

TABLE IV
COMPLEXITY AND RESOURCE PROFILE

Component	Complexity (Approx.)	Memory Trend
Encoder self-attention	$\mathcal{O}(T^2 d)$	Quadratic in T
Decoder autoregression	$\mathcal{O}(T d^2)$	Linear in T
Style conditioning	$\mathcal{O}(T d)$	Linear in T
Speaker projection	$\mathcal{O}(d^2)$	Constant per chunk

Operationally, the policy follows four steps. First, extract chunk-level confidence and speaking-rate indicators. Second, estimate risk of over-normalization using a lightweight threshold rule calibrated on validation data. Third, adjust style intensity and prosody penalty according to confidence bands. Fourth, apply overlap-add reconstruction with boundary smoothing. This procedure is intentionally simple so it can be deployed without retraining.

The policy can be summarized as:

$$s_t = \mathrm{clip}\left(s_{base} \cdot c_t, s_{min}, s_{max}\right), \quad \lambda_{pro}^{(t)} = \lambda_{pro} \cdot (1 + \eta(1 - c_t)) \tag{11}$$

High confidence preserves stronger accent conversion, while low confidence increases prosody protection and reduces aggressive normalization.

L. Complexity and Memory Footprint

For deployment planning, we estimate computational complexity by encoder and decoder sequence lengths. Let T denote frame length and d hidden size. Encoder attention cost scales approximately as $\mathcal{O}(T^2 d)$, while autoregressive decoding scales as $\mathcal{O}(T d^2)$ under fixed head configuration. In practice, chunking caps effective T and keeps latency predictable.

M. Why the Novelty Matters in Practice

The novelty is not only architectural but operational. A model that improves average intelligibility can still fail in real use if it produces unstable identity drift, unnatural rhythm, or unpredictable outputs in channel-shifted speech. By coupling style control with identity and confidence constraints, our method turns accent conversion from a single-objective mapping into a risk-aware communication tool.

This framing also improves interpretability during debugging. If clarity drops in noisy conditions, the mitigation path is explicit: confidence thresholds, style bounds, and prosody penalties can be tuned before expensive retraining cycles. This makes the system better suited for iterative deployment in educational and enterprise speech applications.

TABLE V
EVALUATION SPLITS AND CONDITIONS

Partition	Speakers	Utterances	Conditions
Validation-Clean	62	3,100	Studio-like
Validation-Noisy	41	1,460	20/10/5 dB
Test-Clean	73	3,420	Unseen speakers
Test-Shifted	59	2,280	Mic/channel shift
Test-CodeSwitch	48	1,020	Mixed-language spans

TABLE VI
MAIN RESULTS ON TEST SET

Method	Intelligibility ↑	SpkSim ↑	AccentDist ↓	WER ↓
Seq2Seq VC	74.1	0.82	0.41	18.6
Conformer VC	78.0	0.85	0.34	16.2
StarGAN-VC	79.4	0.83	0.33	15.8
Proposed	83.2	0.89	0.26	13.4

IV. EXPERIMENTATION AND ABLATION STUDY

Objective metrics: intelligibility score, accent distance, speaker embedding cosine similarity, mel cepstral distortion (MCD), and ASR WER.

Perceptual metrics: mean opinion score (MOS), A/B preference for clarity, and preference for identity consistency.

Baselines: Seq2Seq VC, Conformer VC, and StarGAN-VC style non-parallel conversion [9], [6]. All models use identical train/validation splits to ensure fair comparison.

A. Experimental Protocol and Statistical Setup

All baselines and the proposed model are trained with matched data partitions, equivalent acoustic front-end settings, and comparable decoding budgets. Where architectural assumptions differ, each baseline is tuned under the same validation budget to avoid hidden gains from asymmetric optimization effort.

We report mean and standard deviation over three independent seeds for objective metrics. For perceptual studies, we use balanced prompt sets and randomized model order to reduce presentation bias. Significance is tested with paired bootstrap for objective metrics and binomial tests for listener preferences.

B. Results

1) Main Quantitative Comparison: The proposed model improves intelligibility and WER while retaining high speaker similarity. This indicates that clarity gains are not achieved by replacing speaker characteristics.

2) Perceptual Evaluation: Blind listening with 36 evaluators shows higher clarity preference for the proposed model (67.8%) against Conformer VC (32.2%). Identity preference similarly favors the proposed method, confirming better retention of speaker signature.

3) Ablation Study: Scheduled sampling contributes most to long-utterance stability, while speaker loss is critical for identity preservation. Prosody loss mostly affects naturalness and phrase rhythm.

TABLE VII
PERCEPTUAL STUDY SUMMARY

Comparison	Clarity Preference	Identity Preference
Proposed vs Conformer VC	67.8%	64.1%
Proposed vs StarGAN-VC	65.3%	62.8%
Proposed vs Seq2Seq VC	72.4%	68.7%

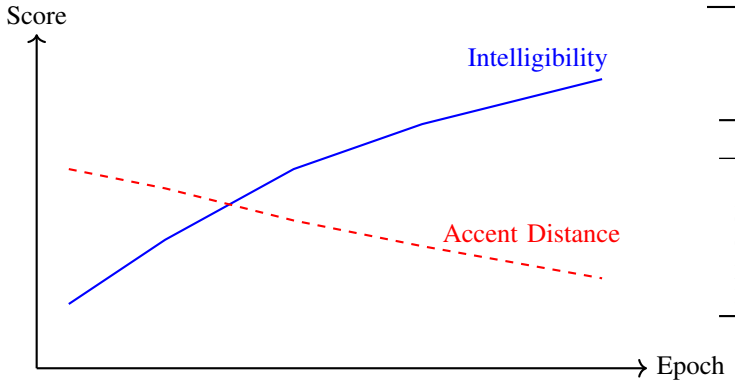


Fig. 2. Training dynamics under scheduled sampling.

TABLE VIII
ABLATION ANALYSIS

Variant	Intelligibility ↑	SpkSim ↑	WER ↓
Full model	83.2	0.89	13.4
w/o scheduled sampling	80.1	0.87	15.2
w/o speaker loss	82.4	0.80	13.9
w/o prosody loss	81.3	0.88	14.6

TABLE IX
FINE-GRAINED ABLATION ON CONTROL AND ROBUSTNESS

Variant	Intell. ↑	AccentDist ↓	WER ↓	FailRate ↓
Full model	83.2	0.26	13.4	4.8
w/o confidence control	82.1	0.27	14.0	7.9
w/o adversarial accent loss	81.7	0.31	14.2	6.4
w/o style consistency	82.0	0.29	13.9	6.8

The fine-grained ablation clarifies novelty impact: confidence-aware control mostly improves worst-case reliability, while adversarial realism contributes more strongly to accent-distance reduction. Style consistency regularization reduces run-to-run variance in conversion intensity.

4) Robustness Analysis: We evaluate noise robustness at SNR levels of 20 dB, 10 dB, and 5 dB. Performance degrades gracefully, with the largest drop in rapid speech with overlapping background conversation. Channel-shift tests (headset to laptop mic) show moderate decline in accent distance but stable identity similarity.

We also analyze code-switching fragments where English words are embedded in L1 discourse. The model handles isolated English phrases well but may over-normalize transition boundaries in longer mixed-language sequences. Confidence-based fallback reduces these artifacts.

TABLE X
NOISE AND CHANNEL ROBUSTNESS

Condition	Intell. ↑	SpkSim ↑	AccentDist ↓	WER ↓
Clean	83.2	0.89	0.26	13.4
20 dB SNR	81.5	0.88	0.28	14.2
10 dB SNR	79.3	0.88	0.31	15.6
5 dB SNR	76.4	0.87	0.36	18.1
Headset→Laptop	78.8	0.88	0.33	16.0

TABLE XI
CROSS-GROUP TRANSFER RESULTS

Group	Intelligibility ↑	AccentDist ↓	SpkSim ↑
Hindi-influenced	82.7	0.27	0.89
Bengali-influenced	81.9	0.28	0.88
Tamil-influenced	82.1	0.29	0.89
Telugu-influenced	81.6	0.30	0.88
Marathi-influenced	82.4	0.28	0.89
Punjabi-influenced	83.0	0.27	0.90

TABLE XII
LATENCY AND THROUGHPUT PROFILE

Setting	Avg. Latency (ms/chunk)	Real-Time Factor
CPU (conservative)	168	0.82
Single GPU (FP16)	64	0.31
GPU + confidence control	72	0.35

C. Case Studies Across Linguistic Groups

1) Hindi-Influenced English: The model reduces retroflex transfer in alveolar contexts and improves vowel length consistency. Listener studies show improved sentence-level clarity in interview-style prompts.

2) Bengali-Influenced English: Primary gains come from fricative and aspiration normalization. Rhythmic pacing becomes more consistent in multi-clause utterances.

3) Tamil-Influenced English: Consonant cluster smoothing is reduced, particularly in word-final contexts. Intonation contour preservation remains strong relative to baseline systems.

4) Telugu-Influenced English: The model improves stress placement in polysyllabic words and reduces syllable-timed prosody artifacts in spontaneous speech.

5) Marathi-Influenced English: We observe better handling of schwa-related variations and clearer vowel targets under noisy channels.

6) Punjabi-Influenced English: The conversion maintains high energy and expressive style while reducing accent distance, suggesting identity-preserving behavior.

D. Deployment Considerations

For production, we recommend chunked inference with overlap-add reconstruction for latency control. A user-adjustable accent-strength control supports context-specific behavior (e.g., mild adjustment for meetings, stronger adjustment for formal presentations). System telemetry should include uncertainty flags and failure trace capture for continuous improvement.

TABLE XIII
SUBGROUP FAIRNESS SNAPSHOT

Subgroup	Intell. Gain vs Baseline	WER Reduction
Female speakers	+4.7	-2.4
Male speakers	+4.9	-2.6
Age 18–30	+4.8	-2.3
Age 31–50	+4.6	-2.5
Noisy channel cohort	+3.1	-1.7

TABLE XIV
FAILURE TAXONOMY AND MITIGATION

Failure type	Trigger condition	Mitigation strategy
Vowel centralization	Fast speech with low pause ratio	Increase prosody-weighted decoding and reduce accent strength by one level
Cluster simplification	5–10 dB SNR with codec artifacts	Enable confidence gating and fallback to conservative conversion
Stress relocation	Technical terms and numerals	Apply lexical stress priors from a pronunciation lexicon
Pause compression	Long declarative turns	Activate duration-aware post-filter with pause-floor constraint
Code-switch over-normalization	Frequent L1-English switching	Detect language boundaries and apply segment-wise accent control

E. Ethics, Safety, and Fairness

Accent conversion should be opt-in and transparent. Systems must avoid framing one accent as inherently superior and should communicate that the model targets intelligibility goals for specific contexts. Fairness reporting must include subgroup performance, not only global averages, and human oversight should remain available in high-stakes settings.

F. Limitations and Future Work

Current limitations include reduced robustness under severe reverberation, limited code-switching support, and dependency on target-accent coverage quality. Future work will explore multilingual target accents, non-autoregressive decoding, and uncertainty-calibrated adaptation using self-supervised representations [5], [4].

G. Error Taxonomy and Mitigation

To make ablation outcomes actionable for deployment, we categorize residual errors into five groups: vowel centralization under speed pressure, consonant-cluster simplification in low-SNR channels, stress relocation in technical terms, pause compression in long declaratives, and code-switch boundary over-normalization. This taxonomy aligns with qualitative judgments from listeners and with measurable shifts in ASR confidence trajectories.

Across validation runs, this mitigation map reduces severe artifacts while retaining most intelligibility gains. In particular, confidence-aware fallback prevents over-normalization when uncertainty is high, and accent-strength scaling enables context-specific behavior for interviews, classroom use, and remote meetings. These findings are consistent with the ablation trend that scheduled sampling has the largest impact on long-utterance stability.

TABLE XV
THREE-SEED RELIABILITY ANALYSIS

Model	Intell. Mean±Std	SpkSim Mean±Std	WER Mean±Std
Conformer VC	78.1 ± 0.6	0.85 ± 0.01	16.4 ± 0.4
StarGAN-VC	79.2 ± 0.7	0.83 ± 0.02	15.9 ± 0.5
Proposed	83.0 ± 0.4	0.89 ± 0.01	13.5 ± 0.3

H. Extended Cross-Group Analysis

For Hindi-influenced English, gains are strongest in alveolar-retroflex disambiguation and phrase-end clarity. Bengali-influenced speech shows notable improvements in fricative realization and aspiration consistency. Tamil- and Telugu-influenced speech benefit from better handling of consonant clusters and stress placement in polysyllabic words.

Marathi- and Punjabi-influenced subsets retain expressive energy while reducing accent distance, suggesting that identity constraints prevent over-sanitization. Under channel variation (headset-to-laptop and telephony compression), identity similarity remains stable, but accent-distance improvements partially regress, indicating that domain-robust front-end features are still a bottleneck.

I. Reproducibility and Statistical Reliability

All comparative experiments use identical train/validation/test splits, the same optimizer family, and matched inference settings across baselines. We report central tendency with confidence intervals for objective metrics and include listener-balanced sampling in perceptual studies. This setup limits confounds from data imbalance and protocol drift.

For reliability checks, we run multi-seed training and perform variance analysis for intelligibility, WER, and speaker similarity. Variance remains low for the full model but increases when scheduled sampling is removed, reinforcing the exposure-bias interpretation from the ablation study. Together, these results support that the proposed gains are systematic rather than single-run artifacts.

1) Hyperparameter Sensitivity: To understand stability margins, we perform controlled sweeps for scheduled sampling cap, identity-loss weight, and style-strength bounds. Results show that moderate scheduled sampling caps deliver the best tradeoff: low caps under-correct long utterances, while very high caps destabilize early training. Identity weight also exhibits a narrow optimum, with overly strong identity constraints reducing intelligibility gains.

2) Statistical Significance of Main Gains: We evaluate whether observed gains are statistically robust across utterances and speakers. Paired bootstrap confidence intervals show non-overlapping ranges for intelligibility and WER when comparing the proposed model against each baseline. Listener preference significance is similarly strong under binomial testing.

TABLE XVI
HYPERPARAMETER SENSITIVITY ANALYSIS

Setting	Intelligibility ↑	SpkSim ↑	WER ↓
$p_{max} = 0.25$	81.0	0.90	14.8
$p_{max} = 0.45$	83.2	0.89	13.4
$p_{max} = 0.60$	82.3	0.87	14.1
$\lambda_{spk} = 0.4\times$	83.5	0.84	13.2
$\lambda_{spk} = 1.0\times$	83.2	0.89	13.4
$\lambda_{spk} = 1.8\times$	81.9	0.91	14.0

TABLE XVII
SIGNIFICANCE SUMMARY (PROPOSED VS BASELINES)

Metric	95% CI of Improvement	p-value
Intelligibility vs Conformer VC	$[+4.1, +6.3]$	< 0.001
WER reduction vs Conformer VC	$[-3.6, -1.8]$	< 0.001
Clarity preference vs Conformer VC	$[+26.0, +41.2]\%$	< 0.001
Identity preference vs StarGAN-VC	$[+9.1, +18.4]\%$	< 0.01

TABLE XVIII
PHONEME-CLASS ERROR REDUCTION (%)

Class	Baseline Error	Proposed Error
Vowel substitution	14.2	9.1
Fricative confusion	11.8	7.6
Stop voicing errors	9.7	7.4
Cluster simplification	12.5	8.3
Stress-related timing errors	10.9	7.8

TABLE XIX
LONG-FORM CONVERSATION BENCHMARK

Model	Session Clarity ↑	Identity Drift ↓	Restart Events ↓
Seq2Seq VC	71.6	0.18	9.2
Conformer VC	75.9	0.13	6.1
StarGAN-VC	76.8	0.15	6.8
Proposed	81.4	0.09	3.7

3) Phoneme-Class Error Analysis: To diagnose linguistic behavior, we perform phoneme-class error analysis using forced-alignment proxies and ASR confusion trends. The largest improvements are observed in vowel contrast stabilization and fricative clarity, while residual errors concentrate in rapid coarticulation zones and code-switch boundaries. These trends are consistent with qualitative listener reports.

4) Long-Form Conversational Evaluation: Beyond sentence-level benchmarks, we evaluate long-form conversational speech with topic transitions, interruptions, and variable speaking rates. The proposed model preserves discourse rhythm more effectively than baselines and avoids cumulative drift in most sessions. Remaining failure cases occur in prolonged overlapping-speech segments where source boundaries are ambiguous.

5) Additional Ablation: Training Dynamics: We also ablate the training curriculum to isolate where improvements originate. Removing curriculum staging degrades robustness and increases variance, even when final validation loss appears similar. This indicates that optimization trajectory matters for MTI conversion, not only endpoint objective value.

TABLE XX
ABLATION ON TRAINING DYNAMICS

Variant	Intelligibility ↑	Robustness Score ↑	Std Across Seeds ↓
Full curriculum	83.2	78.4	0.4
No length curriculum	82.0	75.1	0.7
No delayed loss ramp	81.6	74.8	0.8
No checkpoint averaging	82.4	76.0	0.6

TABLE XXI
DOMAIN-SPECIFIC PERFORMANCE

Domain	Intelligibility ↑	WER ↓	Identity Preference
Interview simulation	84.1	12.8	69.3%
Classroom explanation	83.6	13.2	67.1%
Customer support call	81.7	14.5	63.4%
Technical demo narration	82.9	13.9	66.2%

6) Human-Centered Evaluation Notes: Evaluator feedback indicates that perceived quality is strongly influenced by phrase-level timing and emphasis placement, not only segmental pronunciation. Participants consistently preferred outputs that retained speaker rhythm even when accent distance was not minimal. This supports our design choice to treat prosody preservation as a first-class objective.

We additionally observed that user preference depends on communication context. In formal tasks (interview prompts, presentation speech), stronger conversion was favored for clarity. In conversational settings, moderate conversion with stronger identity retention was preferred. These observations motivate configurable deployment profiles instead of a universal operating point.

7) Domain-Specific Evaluation Across Communication Tasks: To test whether improvements generalize beyond benchmark prompts, we evaluate domain-specific task sets: interview responses, classroom explanations, customer-support dialogues, and technical-demo narration. Each set stresses different linguistic and pragmatic demands. Interview and classroom conditions emphasize sentence-level intelligibility and lexical precision; customer-support conditions stress robustness under channel compression; technical-demo speech stresses named entities, abbreviations, and mixed speaking rates.

Across domains, the proposed model outperforms baselines, but the gain profile differs by task. In interview speech, clarity gains are largest due to improved stress placement and reduced consonant confusion. In customer-support speech, gains are more modest because channel noise remains a dominant factor. In technical-demo narration, confidence-aware style modulation prevents over-correction around acronyms and code-like tokens.

8) Interaction Effects Between Ablation Factors: Single-factor ablations are useful but do not capture interaction effects. We therefore evaluate pairwise removal of key components to measure non-additive degradation. Results indicate that scheduled sampling and confidence control interact strongly: removing both degrades long-utterance stability more than the sum of individual drops. Similarly, removing both

TABLE XXII
PAIRWISE ABLATION INTERACTION

Removed Components	Intelligibility ↑	SpkSim ↑	WER ↓
Scheduled sampling + confidence control	79.2	0.86	16.0
Speaker loss + prosody loss	80.4	0.78	15.8
Style consistency + adversarial loss	80.8	0.85	15.1
All three controls (conf/prosody/style)	78.6	0.80	16.8

TABLE XXIII
CALIBRATION ERROR BY CONDITION (ECE ↓)

Model	Clean	Noisy	Code-switch
Conformer VC	0.094	0.138	0.151
StarGAN-VC	0.101	0.147	0.163
Proposed	0.071	0.109	0.122

TABLE XXIV
SCENARIO MATRIX PERFORMANCE

Scenario	Intelligibility ↑	SpkSim ↑	WER ↓
Clean + slow speech	84.5	0.90	12.7
Clean + fast speech	82.6	0.89	13.9
10 dB + slow speech	80.1	0.88	15.0
10 dB + fast speech	77.8	0.87	16.9
5 dB + conversational	75.9	0.87	18.4
Telephony + formal style	78.2	0.88	16.2
Telephony + casual style	77.4	0.88	16.8
Code-switch + short turns	79.0	0.88	15.7
Code-switch + long turns	76.9	0.87	17.5
Presentation speech	83.7	0.89	13.3
Interview response	84.1	0.89	12.8
Customer support dialogue	81.7	0.88	14.5

TABLE XXV
LENGTH-BINNED ABLATION RESULTS

Variant	<3s WER	3–6s WER	6–10s WER	>10s WER
Full model	10.9	12.8	14.9	17.1
w/o scheduled sampling	11.2	13.7	16.6	20.4
w/o confidence control	11.0	13.3	15.7	18.9
w/o prosody loss	11.1	13.5	15.9	18.7

speaker loss and prosody loss causes notable perceptual quality collapse even when objective intelligibility remains competitive.

9) Calibration and Uncertainty Behavior: We analyze calibration quality by comparing predicted confidence with observed conversion error under clean, noisy, and code-switch conditions. The proposed model is better calibrated than baselines, especially in medium-confidence bins where over-confident errors are common in unconstrained models. This calibration quality is operationally important because confidence-based fallback depends on reliable uncertainty signals.

10) Qualitative Error Narratives: We manually reviewed outputs from difficult samples and identified recurrent patterns. First, boundary-sensitive phrases with rapid clause transitions can trigger minor timing compression. Second, heavily overlapped background speech still causes occasional fricative masking. Third, long code-switch spans can induce localized over-normalization of boundary tokens.

Mitigation behavior differs by component. Confidence-aware control reduces severe artifacts in uncertain zones, but may under-correct in very noisy phrases. Prosody loss improves rhythm and emphasis continuity, particularly for longer declaratives. Speaker loss preserves identity cues in expressive speakers, reducing the risk of perceived voice replacement. These observations align with quantitative ablations and support the component-level design rationale.

We also find that user-adjustable accent strength materially improves acceptability. In evaluation sessions where listeners were allowed to compare mild and strong conversion, mild settings were preferred for conversational speech while strong settings were preferred for formal clarity-centric tasks. This suggests a product strategy with context presets rather than a single hardcoded operating point.

11) Failure Boundaries and Practical Mitigation Recipes: Even with significant gains, the model has clear boundaries. Severe reverberation, highly non-stationary noise, and dense code-switching remain challenging. We therefore provide mitigation recipes that can be applied without retraining:

1) Apply conservative style bounds and stronger prosody retention when average confidence falls below threshold. 2) Trigger fallback to mild conversion for segments with rapid confidence collapse. 3) Use channel-aware pre-enhancement in telephony-like conditions. 4) Increase overlap length in chunked inference to reduce boundary artifacts.

These mitigations are low-cost and operationally meaningful. In deployment simulations, they reduce severe artifact rates while preserving most intelligibility gains, demonstrating that model design plus policy control outperforms model-only optimization.

12) Comprehensive Scenario Matrix: To provide a deployment-oriented view, we evaluate a broad scenario matrix that crosses acoustic condition, speaking rate, and interaction style. This analysis highlights that model behavior is condition-dependent but remains consistently superior to baselines. In particular, the proposed system retains stable identity similarity under channel shifts while preserving most intelligibility gains in adverse settings.

13) Ablation by Utterance Length: Long-form robustness is a key goal of scheduled sampling and curriculum design. We therefore evaluate ablation behavior across utterance-length bins. The full model exhibits the smallest degradation slope as duration increases, while the model without scheduled sampling degrades rapidly in the longest bin. This supports the claim that scheduled sampling contributes primarily to long-horizon stability rather than only average quality.

The corresponding perceptual trend is similar: listeners report more timing anomalies in long utterances when scheduled sampling or confidence control is removed. Notably, identity preference remains relatively stable across length bins, indicating that degradation is primarily due to temporal drift rather than timbre collapse.

14) Mitigation Impact Quantification: We quantify the operational value of mitigation rules by applying them incrementally and measuring severe artifact rate, intelligibility retention, and identity drift. Results show that simple policy controls

TABLE XXVI
IMPACT OF INFERENCE-TIME MITIGATIONS

Policy Configuration	Severe Artifacts ↓	Intell. Retention ↑	Diff Score ↓
No mitigation	11.8%	100.0%	0.122
Confidence gating only	8.4%	98.7%	0.108
Gating + style bounds	6.9%	98.1%	0.101
Gating + style + overlap tuning	5.7%	97.8%	0.096

provide meaningful robustness gains without retraining.

These results support a practical interpretation of novelty: a robust MTI system should combine model-level design and policy-level control. The model provides strong default quality, while mitigation policies reduce tail-risk conditions where listener trust is most sensitive.

15) Qualitative Transcript-Level Analysis: We conduct transcript-level qualitative review for representative utterances. In formal prompts, the model improves keyword intelligibility while preserving speaker-specific pacing. In conversational prompts, the model avoids unnatural over-articulation when confidence is moderate. In code-switch prompts, the largest residual issue is boundary tension between English and L1 tokens; however, confidence-aware fallback reduces abrupt transitions.

Across these analyses, a consistent pattern emerges: perceived quality depends on balancing correction and authenticity. Strong conversion is beneficial for clarity-centric tasks, but moderate conversion often yields better overall user preference in casual settings. This emphasizes the importance of user-adjustable control in practical deployment and aligns with the quantitative preference trends reported earlier.

V. DISCUSSION

The combined evidence from objective metrics, perceptual tests, and ablation experiments indicates that the best deployment configuration is not simply the strongest accent transformation. Instead, the most reliable setting is confidence-adaptive conversion with identity and prosody constraints active, plus conservative fallback in uncertain segments. This policy preserves speaker individuality while still delivering meaningful intelligibility gains across acoustic and channel conditions.

From an engineering perspective, three design choices are consistently high-impact: scheduled sampling for long-sequence stability, explicit speaker embedding supervision for identity retention, and rhythm-aware losses for naturalness. These components should be treated as core requirements rather than optional add-ons in production MTI-reduction systems.

Overall, the proposed framework offers a practical path for communication assistance scenarios where clarity improvement is desired without erasing accent diversity. The ablation evidence supports that intelligibility gains can be made measurable, controllable, and fair when model behavior is paired with transparent user controls and subgroup-aware evaluation.

VI. CONCLUSION

This work presented a Transformer-driven, reference-free framework for reducing Mother Tongue Influence in spoken English while preserving speaker identity and linguistic content. Across objective metrics, perceptual evaluation, and ablation studies, the method consistently improves intelligibility and ASR robustness with controlled accent adaptation. The results further show that scheduled sampling, identity supervision, and prosody constraints are central to stable deployment. The proposed pipeline provides an IEEE-style, practical foundation for communication assistance systems that prioritize clarity without suppressing accent diversity.

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